

Chapter 1: In-Class Exercises

1.3

Problem: An investment of \$1,000 grows by a constant amount of \$250 each year for five years. What does the graph of the amount function $A_{1,000}(t)$ look like if interest is paid continuously using the linear relationship $A_{1,000}(t) = \$1,000 + \$250t$? How about if interest is only paid at the end of each year?

1.4

Problem: Tonya loans Renu \$1,600. Renu promises that in return, she will pay Tonya \$2,000 at the end of four years. To what rate of simple interest does this correspond?

1.4

Problem: Antonio loans his brother Bob \$2,400 for three years at 5% simple interest. The brothers agree that if Bob wishes to repay the loan early, he may do so, and the repayment amount will still be based on 5% simple interest. Find the amount Bob would be required to pay if he makes his repayment at the end of three years. What if the repayment is after two years or after one year? Calculate i_1 , i_2 , and i_3 if the loan lasts the full three years.

1.4

Problem: Brad borrows \$5,000 from Julio on October 14, 1998 at 8% exact simple interest and agrees to repay the loan on May 7, 1999. What is the amount of Brad's required May repayment?

1.4

Problem: Suppose that the loan of Example (1.4.4) is made at 8% ordinary simple interest instead of 8% exact simple interest. What is the amount of Brad's required May repayment?

1.4

Problem: Suppose that the loan of Example (1.4.4) is made by 8% simple interest computed using the Banker's rule instead of 8% exact simple interest. What is the amount of Brad's required May repayment?

1.5

Problem: Fernando deposits \$12,000 in an account at Victory Bank where accounts grow according to the compound interest accumulation function $a(t) = (1.05)^t$ for all nonnegative t . He makes no further deposits or withdrawals until he closes the account six and a half years after he opened it. How much money does Fernando receive when he closes the account?

1.5

Problem: Sarafina also had \$12,000 to deposit. She invested her money at Simpler Bank where accounts grow according to the accumulation function

$$a(t) = (1.05)^{\lfloor t \rfloor} (1 + .05(t - \lfloor t \rfloor)).$$

She also closes her balance after six and a half years. Compare her closing balance to the closing balance Fernando received in Example (1.5.4).

1.5

Problem: Patriot Bank offers a “bracketed account” with compound interest at an annual effective interest rate of 2% on balances of less than \$2,000, 3% on balances of at least \$2,000 but less than \$5,000, and 4% on balances of at least \$5,000. Moises opens an account with \$1,800. Determine the amount function $A(t) = A_{1,800}(t)$ and show that the *function* $A_{1,800}(t)$ is different from the *function* $1,800a(t)$, although their values agree for t in $[0, 5.32]$.

1.6 **Problem:** Chan borrows \$1,000 at a discount rate of 7%. How much extra money does he have the use of?

1.6 **Problem:** Suppose that the growth of money is governed by the accumulation function $a(t) = (1.05)^{\frac{t}{2}}(1 + .025t)$. Find d_4 and i_4 .

1.6 **Problem:** Suppose that an account is governed by compound interest at an annual effective interest rate of 8%. Find an expression for d_n , the discount rate for the n -th year.

1.7

Problem: Suppose that the growth of money for the next five years is governed by simple interest at 5%. How much money should you invest now in order that you have a balance of \$23,000 three years from now?



$$X = 23000 v(3)$$

$$X = 23000 \left(\frac{1}{1 + .05(3)} \right) = 20,000$$

or

$$X a(3) = 23000 \Rightarrow X = 20,000$$

Problem: Again, suppose that the growth of money for the next five years is governed by the linear accumulation function $a(t) = 1 + .05t$ so that

1.7

$$v(t) = \frac{1}{a(t)} = \frac{1}{1 + (.05)t}.$$

If you wish to invest \$10,000 now and an additional \$X two years from today so as to accumulate a total of \$23,000 five years from now, what is the value of X?



equal at $t=5$ Not = $a(5)$ unless Comp. Interest

$$10,000 a(5) + X v(2) a(5) = 23000$$

$$10,000 (1 + .05(5)) + \frac{X (1 + .05(5))}{1 + .05(2)} = 23000$$

$$\Rightarrow X = 9240$$

equal at $t=0$

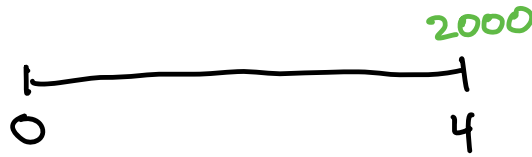
$$10,000 + X v(2) = 23000 v(5)$$

$$10,000 + X \left(\frac{1}{1 + .05(2)} \right) = 23000 \left(\frac{1}{1 + .05(5)} \right) \Rightarrow X = 9240$$

or:

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Problem: What is the present value of \$2,000 to be paid in four years assuming money grows by compound interest at an annual effective interest rate of 6%? What if money grows by compound interest at an annual effective interest rate of 3%?

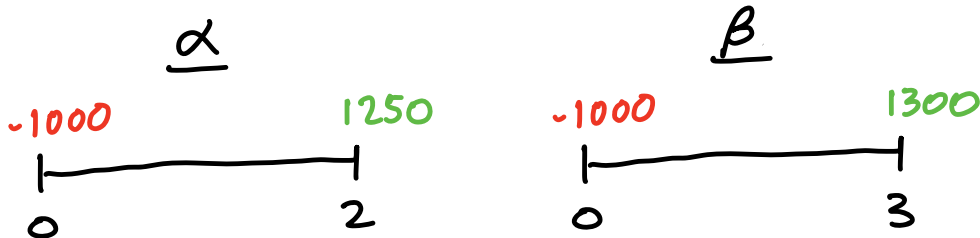


$$PV(\text{when } i=6\%) = 2000 v(4) = 2000 (1.06)^{-4} = 1584.19$$

$$PV(\text{when } i=3\%) = 2000 v(4) = 2000 (1.03)^{-4} = 1776.97$$

1.7

Problem: Compare the net present values of two certificates of deposit available to Helga, one at Bank Alpha with purchase price \$1,000 and redemption value \$1,250 at the end of two years, the other at Bank Beta with purchase price \$1,000 and redemption value \$1,300 at the end of three years, first using the compound interest accumulation function $a(t) = (1.05)^t$ and next using the compound interest accumulation function $a(t) = (1.03)^t$.



$$PV(\alpha \text{ at } 5\%) = 1250 v(2) + -1000 = 1250 (1.05)^{-2} - 1000 = 133.7868$$

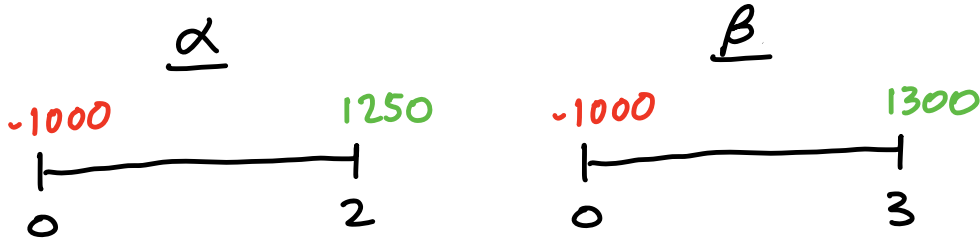
$$PV(\beta \text{ at } 5\%) = 1300 v(3) + -1000 = 1300 (1.05)^{-3} - 1000 = 122.9889$$

$$PV(\alpha \text{ at } 3\%) = 1250 v(2) + -1000 = 1250 (1.03)^{-2} - 1000 = 178.2449$$

$$PV(\beta \text{ at } 3\%) = 1300 v(3) + -1000 = 1300 (1.03)^{-3} - 1000 = 189.6842$$

1.7

Problem: As in Example (1.7.9), Helga has \$1,000 to invest. She has a choice of two investments. Bank Alpha offers a two-year certificate of deposit in which her \$1,000 would grow to \$1,250. Bank Beta's three-year certificate of deposit would allow her \$1,000 to grow to \$1,300. Find the annual effective rate of compound interest to which each of these investments correspond.

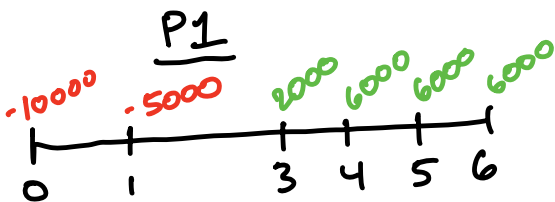


$$i_\alpha \rightarrow 1000 a(2) = 1250 \Rightarrow 1000(1+i_\alpha)^2 = 1250 \Rightarrow i_\alpha = 11.8\%$$

$$i_\beta \rightarrow 1000 a(3) = 1300 \Rightarrow 1000(1+i_\beta)^3 = 1300 \Rightarrow i_\beta = 9.139\%$$

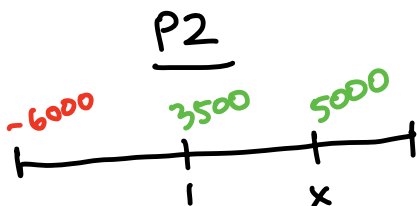
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Problem: Project 1 requires an investment of \$10,000 at time 0 and an additional investment of \$5,000 at $t = 1$. It returns \$2,000 at $t = 3$ and \$6,000 at times $t = 4$, $t = 5$, and $t = 6$. Project 2 requires an investment of \$6,000 at $t = 0$ and returns \$3,500 at $t = 1$ and \$5,000 at an unknown time. The net present values of the two projects are equal when calculated using compound interest at 4%. Find the unknown time for the second return of Project 2.



$$\text{Note: } v(t) = (1.04)^{-t}$$

$$\begin{aligned} PV(P_1) &= -10000 + -5000v(1) + 2000v(3) \\ &\quad + 6000v(4) + 6000v(5) + 6000v(6) \\ &= 1772.575351 \end{aligned}$$



$$\begin{aligned} PV(P_2) &= -6000 + 3500v(1) + 5000v(x) \\ &= -2634.615385 + 5000(1.04)^{-x} \end{aligned}$$

$$\text{Setting } PV(P_1) = PV(P_2) \rightarrow x = -\ln\left(\frac{1772.57... + 2634}{5000}\right) / \ln(1.04)$$

$$= 3.2176989$$

1.9

Problem: Given that the annual effective discount rate is 4.386286%, compute the annual effective interest rate i as a percent.

$$i = \frac{d}{1-d} \Rightarrow i = \frac{.04386286}{.95613714} = .045875072$$

1.9

Problem: Cassandra needs to borrow money to pay her tuition. She has a choice of borrowing at an annual effective interest rate of 5.1% or at an annual effective discount rate of 4.9%. Which rate should she choose?

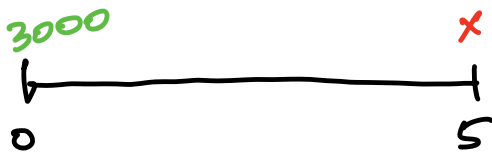
$$i = 5.1\% \Rightarrow d = \frac{.051}{1.051} = .048525214 < .049$$

better to borrow at $i = 5.1\%$.

1.9

Problem: Ezra has the opportunity to borrow money according to compound discount at an annual effective discount rate of 8%. He would like to borrow money in order to completely pay for a \$3,000 used piano. Ezra wishes to repay the loan with a payment in exactly five years. How much money must he ask to borrow at 8% discount?

$$d = 8\% \quad (\text{so } i = \frac{.08}{.92} = .086956522)$$



$$3000 a(5) = X \Rightarrow X = 3000 (1 - .08)^{-5}$$

$$\text{or } X = 3000 (1 + .086956522)^5 \quad \left. \vphantom{X = 3000 (1 + .086956522)^5} \right\} = 4551.789$$

1.10

Problem: National Bank advertises a savings account paying 4% nominal interest compounded quarterly. What is the annual effective yield **APY** for this account?

1.10

Problem: Sandra inherits \$10,000. She deposits it in a five-year certificate of deposit paying 6% nominal interest compounded monthly and the interest remains on deposit. At the end of the five years, Sandra decides to renew her CD for another five years at the then current nominal interest rate of 7.5% compounded quarterly. Again, interest is left to accrue. At the time her second CD matures, what is her investment worth?

1.10

Problem: Vladimir deposits \$50,000 in a three-year certificate of deposit for which interest is compounded quarterly and is left to accrue. At the end of the three years, the balance in the CD is \$63,786.11. What is the nominal annual interest rate $i^{(4)}$ convertible quarterly?

1.10

Problem: Jolene invests money in a fund for which interest is paid once every two years. The effective rate per two-year period is 14%. Find the nominal interest rate convertible biennially⁷ and the annual effective interest rate governing the fund.

1.10

Problem: Trust Bank offers a savings account with a nominal discount rate of 4.8% payable monthly. Find the annual effective rate of interest i and the annual effective rate of discount d .

1.10

Problem: Given that $d^{(4)} = 4.6\%$, find $i^{(12)}$.

1.11 **Problem:** Estelle deposits \$12,500 in a four-year certificate of deposit at Community Trust Bank. If the annual percentage yield (APY) is 4.235%, find the nominal interest rate δ convertible continuously.

1.11 **Problem:** Money at Swift National Bank grows at 3.8% compounded continuously. Rafael Ortiz closed his savings account at Swift exactly three years after he opened it, and he received his balance which was \$24,812. Mr. Ortiz made a \$6,500 deposit exactly one year after he opened the account, but he made no other deposits or withdrawals except for an unspecified opening deposit X . Find X and his balance exactly two years after he opened the account.

1.12 **Problem:** Suppose that you have simple interest at a rate r . Find the force of interest δ_t .

1.12 **Problem:** Suppose that you have compound interest at a rate i . Find the force of interest δ_t as a function of i .

1.12 **Problem:** Suppose $a(t) = (1.05)^{\frac{t}{2}} (1.04)^{\frac{t^2}{3}} (1.03)^{\frac{t^3}{6}}$. Find δ_t .

1.12 **Problem:** Suppose $\delta_t = \frac{3}{1-3t}$. Find the corresponding accumulation function $a(t)$.